

PROBLEM STATEMENT

Product of Linear & Power Expressions

Find the derivative with respect to x of the function:

$$y = (3x - 2)(5x + 4)^4$$

STEP 1 প্রদত্ত গুণফলীয় বহুপদী ফাংশনটি উল্লেখ করি (Given Polynomial Product Function)

$$y = (3x - 2)(5x + 4)^4$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating both sides w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} [(3x - 2)(5x + 4)^4]$$

STEP 3 গুণন সূত্র বা Product Rule প্রয়োগ করি ($\frac{d}{dx}(uv) = uv' + vu'$)

$$\frac{dy}{dx} = (3x - 2) \cdot \frac{d}{dx} [(5x + 4)^4] + (5x + 4)^4 \cdot \frac{d}{dx} (3x - 2)$$

STEP 4 চেইন রুল প্রয়োগ করে প্রতিটি অংশের অন্তরক নির্ণয় করি (Evaluating Individual Derivatives)

$$\begin{aligned}\frac{d}{dx} [(5x + 4)^4] &= 4(5x + 4)^3 \cdot \frac{d}{dx} (5x + 4) = 4(5x + 4)^3 \times 5 = 20(5x + 4)^3 \\ \frac{d}{dx} (3x - 2) &= 3\end{aligned}$$

STEP 5 সাধারণ উৎপাদক $(5x + 4)^3$ কমন নিয়ে ব্রাকেটের ভেতরের অংশ সরল করি (Factoring Common Term)

$$\begin{aligned}\frac{dy}{dx} &= (3x - 2) \cdot 20(5x + 4)^3 + 3(5x + 4)^4 = (5x + 4)^3 [20(3x - 2) + 3(5x + 4)] \\ \frac{dy}{dx} &= (5x + 4)^3 [60x - 40 + 15x + 12] = (75x - 28)(5x + 4)^3\end{aligned}$$

FINAL ANSWER $\frac{dy}{dx} = (75x - 28)(5x + 4)^3$

BOX 1: FORMULATION & PRODUCT-POWER RULES

- **General Linear Composite Form:** If $y = (ax + b)(cx + d)^n$, then:

$$\frac{dy}{dx} = a(cx + d)^n + nc(ax + b)(cx + d)^{n-1} = (cx + d)^{n-1}[a(cx + d) + nc(ax + b)]$$

- For $a = 3, b = -2, c = 5, d = 4, n = 4$:

$$a(cx + d) + nc(ax + b) = 3(5x + 4) + 4(5)(3x - 2) = 15x + 12 + 60x - 40 = 75x - 28$$

- Thus, $\frac{dy}{dx} = (75x - 28)(5x + 4)^3$.

BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Dropping the Inner Derivative 5:** Writing $\frac{d}{dx}(5x + 4)^4 = 4(5x + 4)^3$ without multiplying by 5.
- **Arithmetic Slips in Linear Sum:** Errors when simplifying $-40 + 12 = -28$.

BOX 3: REAL DOMAIN & STATIONARY POINTS

- **Domain of Definition:** Entire real line $\mathbb{R} = (-\infty, \infty)$.
- **Critical Points:** $\frac{dy}{dx} = 0$ when:
 1. $5x + 4 = 0 \implies x = -\frac{4}{5} = -0.8$ (triple root, inflection point).
 2. $75x - 28 = 0 \implies x = \frac{28}{75} \approx 0.3733$ (local minimum).

BOX 4: LOGARITHMIC DIFFERENTIATION ALTERNATIVE

- Taking natural logarithm:

$$\ln y = \ln(3x - 2) + 4 \ln(5x + 4)$$

- Differentiating:

$$\frac{1}{y} \frac{dy}{dx} = \frac{3}{3x - 2} + \frac{20}{5x + 4} = \frac{3(5x + 4) + 20(3x - 2)}{(3x - 2)(5x + 4)} = \frac{75x - 28}{(3x - 2)(5x + 4)}$$

- Multiplying by $y = (3x - 2)(5x + 4)^4$:

$$\frac{dy}{dx} = (75x - 28)(5x + 4)^3$$

BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 4 / 5 MARKS)

- **1.0 Mark:** Correct identification and setup of Product Rule $\frac{d}{dx}(uv) = uv' + vu'$.
- **1.0 Mark:** Accurate chain rule computation of $\frac{d}{dx}(5x + 4)^4 = 20(5x + 4)^3$.
- **1.0 Mark:** Factoring out $(5x + 4)^3$.
- **1.0 Mark:** Correct linear algebra simplification to $(75x - 28)(5x + 4)^3$.